0. **Code:** bhagams4063 bhagamsaigoutham.zip

1. NestedLoop/Larger running average/Program.cs
2. NestedLoop/NestedLoop/Program.cs

1. NestedLoop/Larger running average/Program.cs

```
namespace Larger_running_average
{
    internal class Program
    {
        static void Main(string[] args)
        {
            int[] numbers = {
                7,
                3,
                1,
                8,
                2,
                4,
                5,
                0,
                9
            };
            double result = calculateAverage(numbers);
            Console.WriteLine(result);
        }
        static double calculateAverage(int[] array)
        {
            //check if array is not null and array length is not 0
            if (array == null && array.Length == 0)
            {
                return 0;

            }
            double sum = 0;
            int count = 0;
            double prevsum = 0;
```

A1

A3

1. **NestedLoop/Larger running average/Program.cs**

```
        double runningavg = 0;

        for (int i = 0; i < array.Length; i++)
        {
            //check if running avg with the prevsum
            prevsum += array[i];
            runningavg = prevsum / (i + 1);

            //if array[i] > running avg then we have to take that items
              and return their avg
            if (array[i] > runningavg)
            {
                sum += array[i];
                count++;

            }

            if (i == 0)
            {
                runningavg = 0;
            }

        }
        if (count == 0)
        {
            return 0;

        }
```

can be inside the loop since it's not a control variable.

if(i>0) {

(i-1)

check after the first value.

A2c

A2

A2b

Flawed avg. logic, because we can have average being Ø.

A3

1. **NestedLoop/Larger running average/Program.cs**

```
            return sum / count;

        }
    }
}
```

2. NestedLoop/NestedLoop/Program.cs

```
namespace NestedLoop
{
    internal class Program
    {
        static void Main(string[] args)
        {
            int[] orgarray = {
                7,
                3,
                1,
                8,
                2,
                4,
                5,
                0,
                9
            };

            int[] modarray = nested(orgarray);
            Console.WriteLine("\n\n modified array:");
            for (int i = 0; i < modarray.Length; i++)
            {
                Console.Write(modarray[i] + " ");
            }

        }
        //creating method
        static int[] nested(int[] array)
        {
            //creating and returning the new array
            int n = array.Length;
```

2. **NestedLoop/NestedLoop/Program.cs**

```
        int newsize = n * (n + 1) / 2;
        int[] newarray = new int[newsize];
        //copying all orginal array values to new array
        for (int i = 0; i < n; i++)
        {
            newarray[i] = array[i];
        }

        int index = n;
        for (int i = 0; i < n; i++)
        {
            int minindex = -1;
            int minvalue = int.MaxValue;

            //check the previously inserted values with the index
            for (int j = 0; j < n; j++)
            {
                if (array[j] < minvalue)
                {
                    minvalue = array[j];
                    minindex = j;
                }
                if (j == minindex)
                {
                    newarray[index] = array[j];
                    index++;
                }

            }
```

B2

B3

find the smallest value for the entire subarray of the newarray.

always max value

wrong array.

has to be in a separate loop

B4

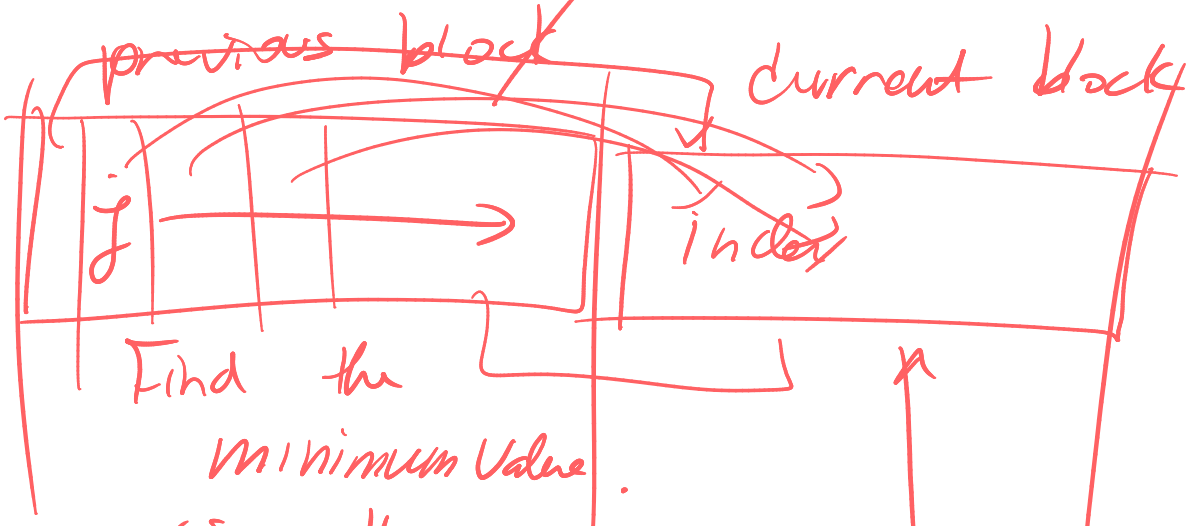
newarray[—]

2. **NestedLoop/NestedLoop/Program.cs**

```
                array[minindex] = int.MaxValue;

            }



            return newarray;
        }
    }
}
```

What?

Copy.



index $(i-1)^{th}$ iteration

i^{th} iteration

need to keep track of additional indexes.